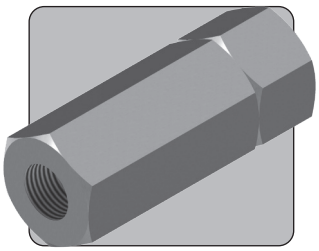
NR50 -12K

CHECK VALVE

1/2"

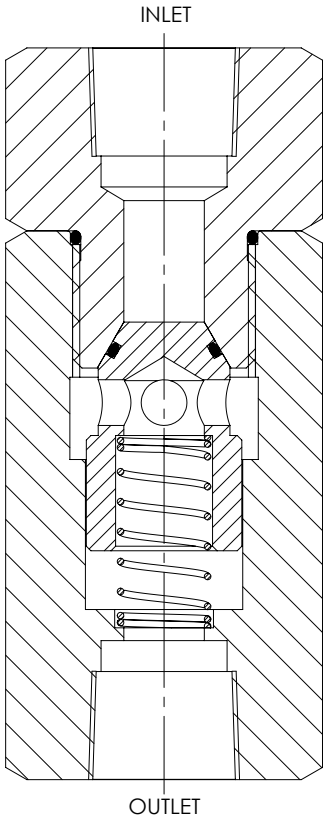
UP TO 828 BAR 12,000 PSI



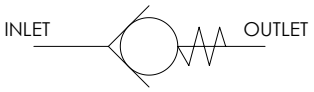
The NR50 is a 1/2" nominal bore check valve. It will allow flow in one direction, and blocks flow in the opposite direction. It may also be referred to as a non-return valve.

The valve uses sealed poppet technology to give virtually zero leakage with liquids and gases.

- Check or Non-Return valve
- Stainless steel construction
- Sealed poppet design
- Virtually zero leakage
- Can be used subsea
- Suitable for liquid or gas
- Suitable for use with air, nitrogen, sweet natural gas, mineral oils, water glycols, water and sea water
- Suitable for many other media, contact us for advice
- NACE MR01-75 compliant version available
- High cracking pressure variants available, contact us for details
- Various seal options available
- Various porting options available



Specifications

BASIC MODEL NUMBER	NR50
SYMBOL	
MAX WORKING PRESSURE (LIQUID)	828 bar (12,000 psi)
MAX WORKING PRESSURE (GAS)	414 bar (6,000 psi)
CV (FLOW CAPACITY)	1.6 - 3.1 See Product Selector opposite
FLUID	Liquids and Gases See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2"
CRACKING PRESSURE	0.1 - 0.3 bar (2 - 5 psi)
WEIGHT	0.9 kg (2.0 lb)

Specifications may change without notice

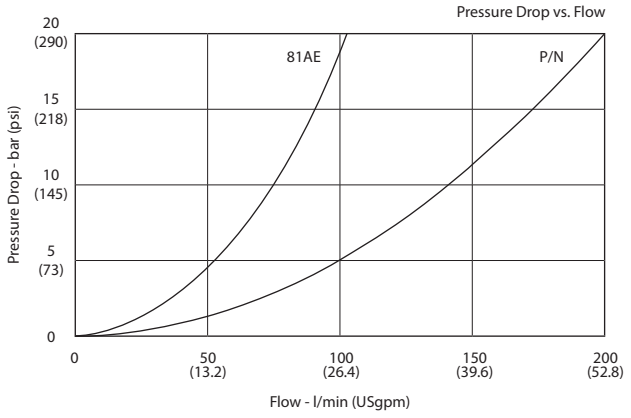
Materials

Externally Exposed Parts: 316 stainless steel.
Internally Wetted Parts: 316 stainless steel with 302 stainless steel spring. The spring is inconel for SWD and NACE variants.

The standard valve is designed for use with air, nitrogen, sweet natural gas, mineral oil, water glycols and plain water, and may also be used with a wide variety of media compatible with the materials of construction.

⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

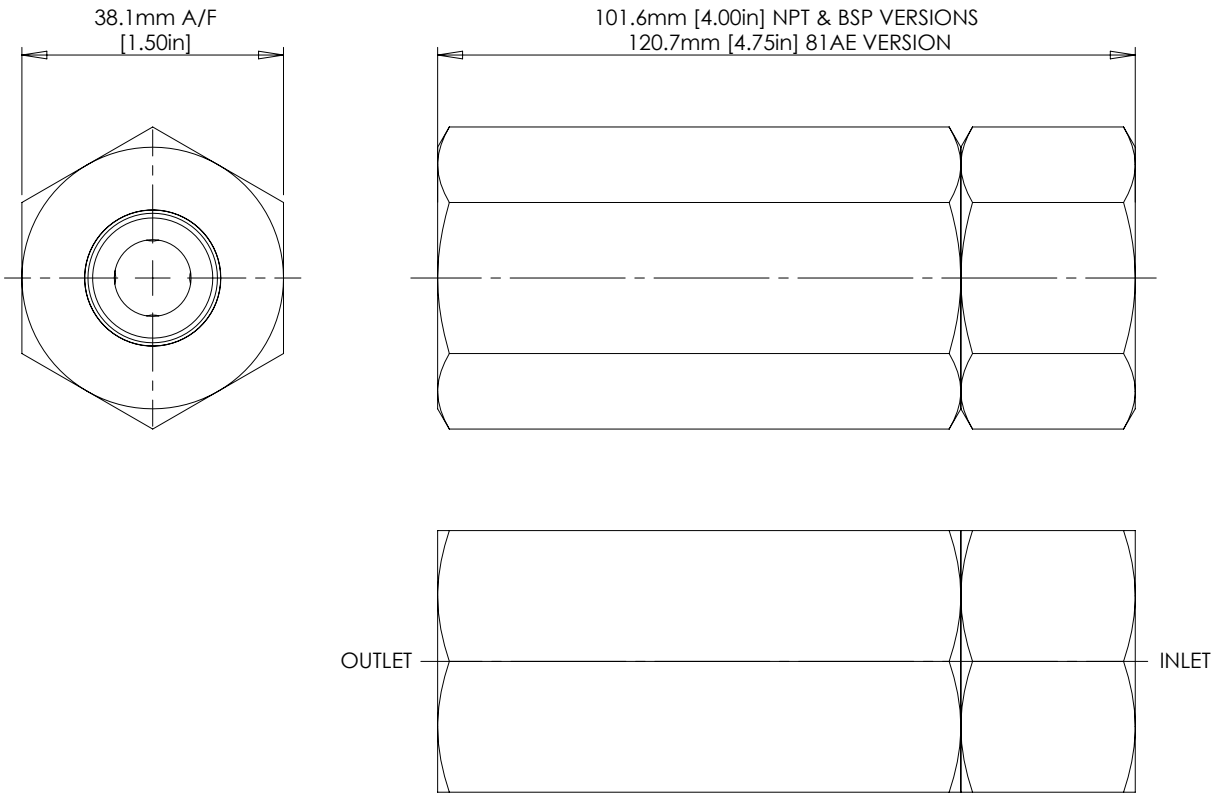
Typical Performance



Typical performance based on water

Installation Diagram





Product Selector

NR50 P - NBR - SWD - 12K

VALVE TYPE	PORTING OPTIONS	SEAL OPTIONS	VALVE OPTIONS	MAX WORKING PRESSURE
NR50	P 1/2" BSPP female 3.1CV	Leave blank for Viton® seals -10˚C to +160˚C	Leave blank if not required	12K 828 bar (12,000 psi) Liquid
	N 1/2" NPT female 3.1CV	NBR Nitrile seals -30˚C to +105˚C	SWD Materials compatible with sea water	414 bar (6,000 psi) Gas
	-81AE 9/16" OD medium pressure tube port 1.6CV	EP EPDM seals -45˚C to +150˚C	NACE Materials comply with NACE MR01-75	
		KLZ Kalrez® seals -25˚C to +200˚C		
		HNBR Hydrogenated Nitrile seals -20˚C to +150˚C		
Repair and seal kits are available on request For a seal kit add SK at the end of the model code and for a repair kit add RK		Further seal options are available on request		